



Ha Huy Hoang

LAB 2

OPENSTACK – APIs

NỘI DUNG

có những gì?

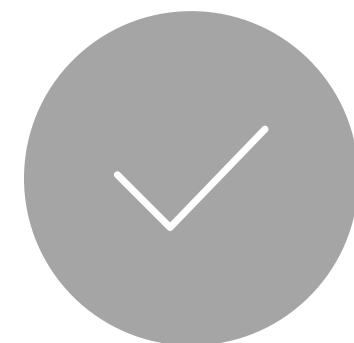
2



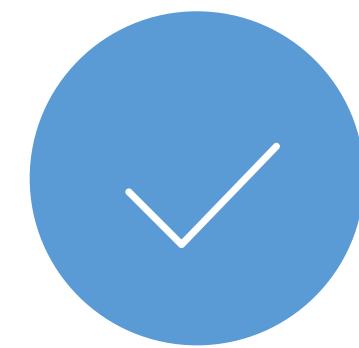
Mục tiêu bài Lab



Giới thiệu Postman



Giới thiệu Openstack APIs



Demo các thao tác cơ bản



Yêu cầu thực hành

HỌC GÌ VÀ LÀM ĐƯỢC GÌ

hôm nay?

3

Authentication



Tạo Network/
Subnets/ Ports



Sử dụng
user_data

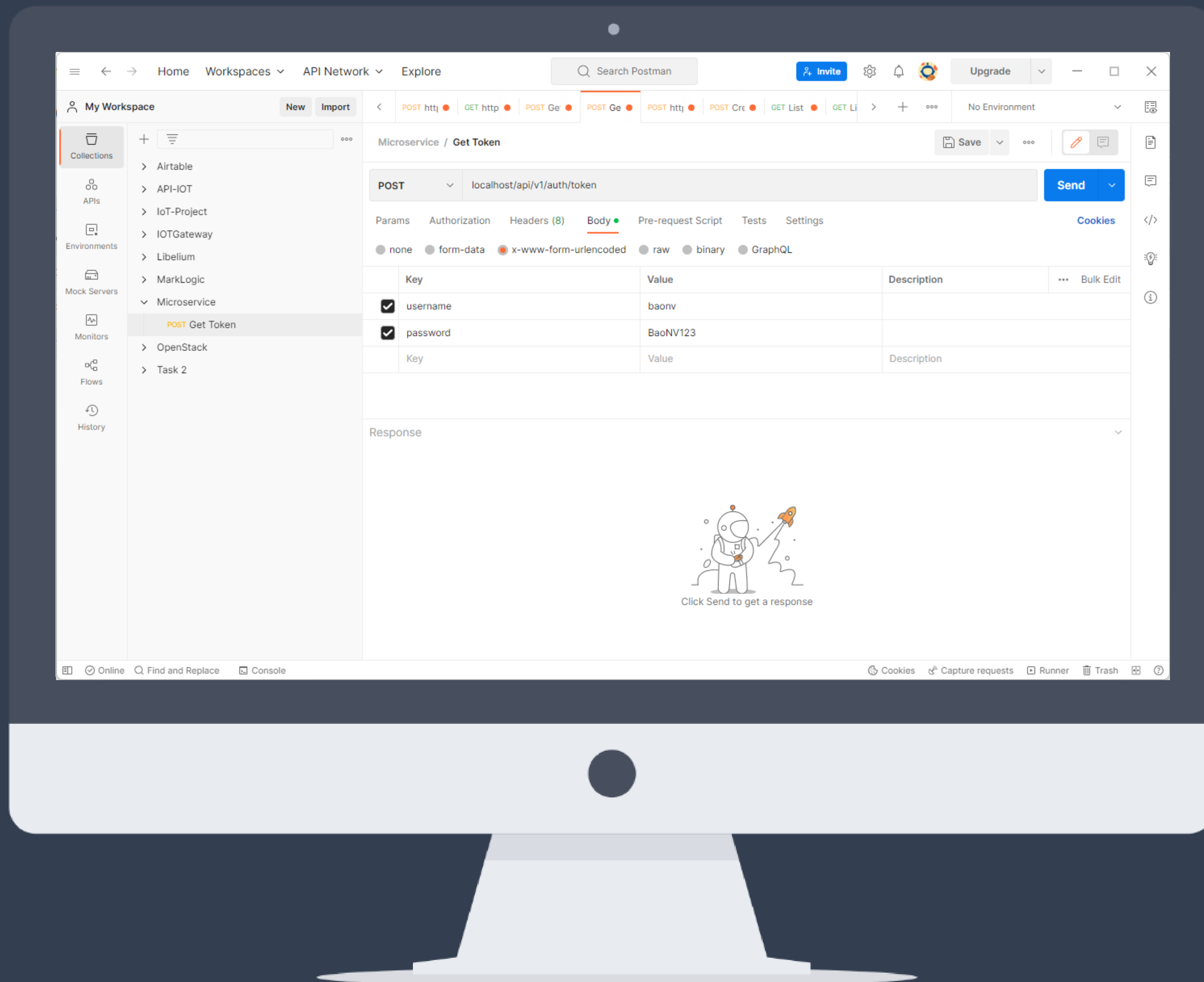


Tạo instance



Triển khai, cấu hình

Hệ thống máy ảo theo
kiến trúc mạng cho trước



Postman

A web developers helper program to create and test custom HTTP requests

Demo trực tiếp tại lớp

GIỚI THIỆU OPENSTACK APIs



GIỚI THIỆU OPENSTACK APIs



<https://developer.openstack.org/api-guide/quick-start/#current-api-versions>

Liên kết đến các Openstack APIs liệt kê
phân nhóm theo projects



<https://developer.openstack.org/api-ref/identity/v3/>

Identity API v3

Các APIs hỗ trợ chứng
thực, quản lý users



<https://developer.openstack.org/api-ref/network/v2/>

Networking API v2.0

Các APIs hỗ trợ quản lý
Networks, Subnets,
Ports, Router, Firewall...



<https://developer.openstack.org/api-ref/image/v2/>

Image Service API v2

Các APIs hỗ trợ quản lý
Images



<https://developer.openstack.org/api-ref/compute/>

Compute API

Các APIs hỗ trợ quản lý
Instance, Flavors

DEMO CÁC THAO TÁC CƠ BẢN

Demo trực tiếp tại lớp



☰

← →

Home

Workspaces ▾

API Network ▾

Explore

🔍 Search Postman

👤 ⚙️ 🔔 🔄

Upgrade ▾

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POST http

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GET List

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GET List

> + ...

No Environment ▾

OpenStack / Get Token

Save ▾ ...

🔗 💬

POST ▾

http://192.168.27.10:5000/v3/auth/tokens

Send ▾

Params

Authorization

Headers (9)

Body ●

Pre-request Script

Tests

Settings

Cookies

Beautiful

● none

● form-data

● x-www-form-urlencoded

● raw

● binary

● GraphQL

JSON ▾

1 {

2 ... "auth": {

3 ... "identity": {

4 ... "methods": [

5 ... "password"

6 ...],

7 ... "password": {

8 ... "user": {

9 ... "name": "gr1",

10 ... "domain": {

11 ... "name": "Default"

12 ... },

13 ... "password": "123"

14 ... }

15 ... }

16 ... }

17 ... }

18 ... }

Body

Cookies

Headers (9)

Test Results

🌐 Status: 201 CREATED Time: 395 ms Size: 3.89 KB Save as Example ...

Content-Length

④

3497

X-Subject-Token

④

gAAAAABkIxiaIAKJw56AgUhN9kiLiR1nctJCbRWk99vRptLKxcZFTh_5JBYmpUKJ7hg...

Vary

④

X-Auth-Token

x-openstack-request-id

④

req-07fbd2a9-8026-42a5-8ff4-8d496f4b32ac

📄

🔍 Online

🔍 Find and Replace

📄 Console

🍪 Cookies

📄 Capture requests

🏃 Runner

🗑️ Trash

🔍

Demo trực tiếp tại lớp

POST ⌵ http://192.168.27.10:9696/v2.0/networks Send ⌵

Params Authorization **Headers (10)** Body ● Pre-request Script Tests Settings Cookies

☒	Accept	ⓘ	*/*	
☒	Accept-Encoding	ⓘ	gzip, deflate, br	
☒	Connection	ⓘ	keep-alive	
☒	Content-Type		application/json	
☒	X-Auth-Token		gAAAAABkIxialAKJw56AgUhN9kiLiR1nctJCbRWk99v...	
	Key		Value	Description

Body Cookies Headers (7) Test Results 🌐 Status: 201 Created Time: 228 ms Size: 832 B 📄 Save as Example ⋮

Pretty Raw Preview Visualize JSON ⌵ 📄 🔍

1 🔗 📄

Để chứng thực cho các API Request ta thêm trường X-Auth-Token vào header với giá trị đã được lấy từ bước lấy token

Demo trực tiếp tại lớp

CHỨNG THỰC

The screenshot shows the Postman interface with a POST request to `http://192.168.27.10:9696/v2.0/networks`. The request body is a JSON object: `{ "network": { "name": "baonv_test", "admin_state_up": true } }`. The response status is `201 Created` with a time of `228 ms` and a size of `832 B`. The response body is a detailed JSON object: `{ "network": { "id": "102cb515-1caf-4b37-b08c-f6478c51d8c0", "name": "baonv_test", "tenant_id": "a069c5716c9d46deb99c9587322fbec9", "admin_state_up": true, "mtu": 1450, "status": "ACTIVE", "subnets": [], "shared": false, "project_id": "a069c5716c9d46deb99c9587322fbec9", "port_security_enabled": true, "router:external": false, "is_default": false, "availability_zone_hints": [], "availability_zones": [], "ipv4_address_scope": null } }`.

OpenStack / Create Network

POST `http://192.168.27.10:9696/v2.0/networks` Send

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies Beautify

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "network": {
3     "name": "baonv_test",
4     "admin_state_up": true
5   }
6 }
```

Body Cookies Headers (7) Test Results Status: 201 Created Time: 228 ms Size: 832 B Save as Example

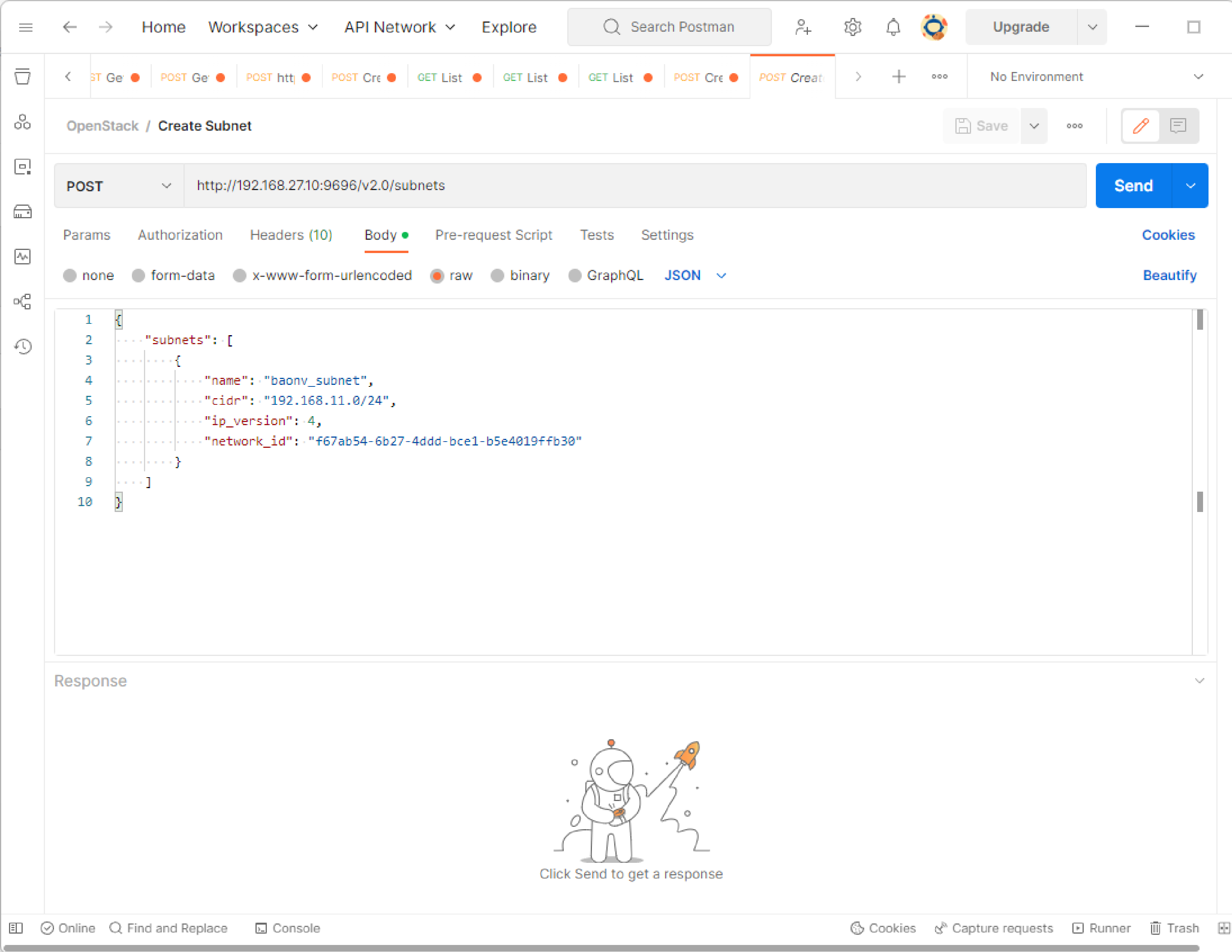
Pretty Raw Preview Visualize JSON

```
1 {
2   "network": {
3     "id": "102cb515-1caf-4b37-b08c-f6478c51d8c0",
4     "name": "baonv_test",
5     "tenant_id": "a069c5716c9d46deb99c9587322fbec9",
6     "admin_state_up": true,
7     "mtu": 1450,
8     "status": "ACTIVE",
9     "subnets": [],
10    "shared": false,
11    "project_id": "a069c5716c9d46deb99c9587322fbec9",
12    "port_security_enabled": true,
13    "router:external": false,
14    "is_default": false,
15    "availability_zone_hints": [],
16    "availability_zones": [],
17    "ipv4_address_scope": null
18  }
19 }
```

Online Find and Replace Console Cookies Capture requests Runner Trash

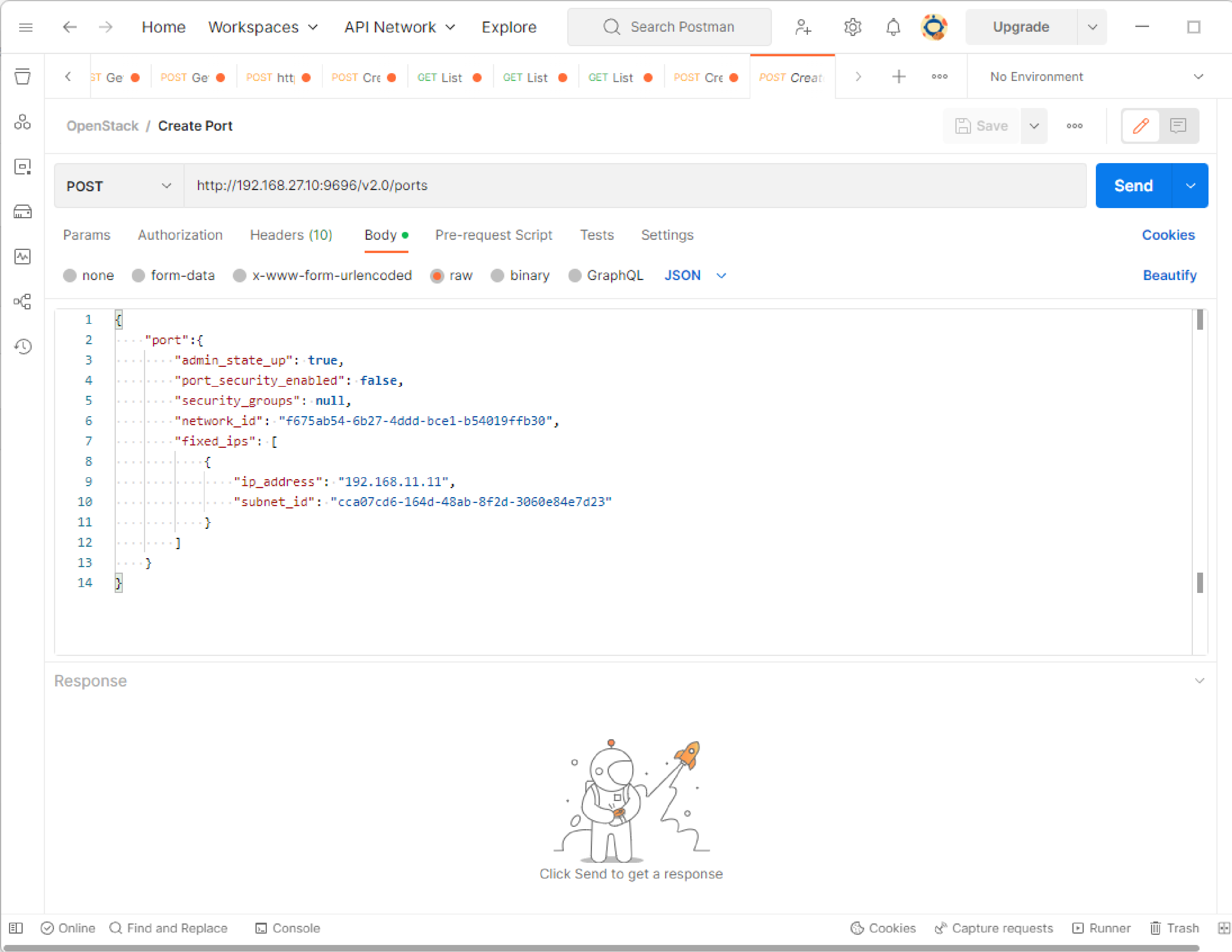
Demo trực tiếp tại lớp

TẠO NETWORK



Demo trực tiếp tại lớp

TẠO SUBNET



Demo trực tiếp tại lớp

TẠO PORT

The screenshot shows the Postman interface with a GET request to `http://192.168.27.10:9292/v2/images`. The request is configured with two headers: `Content-Type` set to `application/json` and `X-Auth-Token` set to `{{openstack-token}}`. The response is displayed in the Body tab, showing a JSON array of image details for "CentOS 9 Stream".

Request Details:

- Method: GET
- URL: `http://192.168.27.10:9292/v2/images`
- Headers (8):
 - `Content-Type`: `application/json`
 - `X-Auth-Token`: `{{openstack-token}}`

Response Details:

- Status: 200 OK
- Time: 97 ms
- Size: 9.01 KB

Response Body (JSON):

```
{
  "images": [
    {
      "owner_specified.openstack.md5": "",
      "owner_specified.openstack.object": "images/CentOS 9 Stream",
      "owner_specified.openstack.sha256": "",
      "name": "CentOS 9 Stream",
      "disk_format": "qcow2",
      "container_format": "bare",
      "visibility": "public",
      "size": 934675456,
      "virtual_size": 10737418240,
      "status": "active",
      "checksum": "fcf46b81cf0ed9291b44960091b3c94d",
      "protected": true,
      "min_ram": 0,
      "min_disk": 0
    }
  ]
}
```

Demo trực tiếp tại lớp

LIST IMAGES

☰

← → Home Workspaces ▾ API Network ▾ Explore

🔍 Search Postman

👤 ⚙️ 🔔 🔄

Upgrade ▾

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OpenStack / List Flavors Save ▾ ...

GET ▾ http://192.168.27.10:8774/v2.1/flavors Send ▾

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

Headers 6 hidden

	Key	Value	Description	...	Bulk Edit	Presets ▾
✓	Content-Type	application/json				
✓	X-Auth-Token	{{openstack-token}}				
	Key	Value	Description			

Body Cookies Headers (11) Test Results

🌐 Status: 200 OK Time: 37 ms Size: 1.57 KB Save as Example ...

Pretty Raw Preview Visualize JSON ▾

```
1 {
2   "flavors": [
3     {
4       "id": "01-01-15",
5       "name": "tiny.1",
6       "links": [
7         {
8           "rel": "self",
9           "href": "http://10.71.2.44/v2.1/flavors/01-01-15"
10        },
11        {
12          "rel": "bookmark",
13          "href": "http://10.71.2.44/flavors/01-01-15"
14        }
15      ]
16    },
17  ]
18 }
```

📄 Online 🔍 Find and Replace 🖨 Console

🍪 Cookies 🔄 Capture requests 🏃 Runner 🗑 Trash 🗑

Demo trực tiếp tại lớp

LIST FLAVORS

HomeWorkspaces ▾API Network ▾Explore

Search Postman

Upgrade

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No Environment

OpenStack / List Keypairs

Save

GET

http://192.168.27.10:8774/v2.1/os-keypairs

Send

ParamsAuthorizationHeaders (8)BodyPre-request ScriptTestsSettingsCookies

Headers

6 hidden

	Key	Value	Description		Bulk Edit	Presets ▾
☒	Content-Type	application/json				
☒	X-Auth-Token	{{openstack-token}}				
	Key	Value	Description			

Response

Click Send to get a response

OnlineFind and ReplaceConsole

CookiesCapture requestsRunnerTrash

Demo trực tiếp tại lớp

LIST KEYPAIRS

☰

HomeWorkspaces ▾API Network ▾Explore

🔍

Search Postman

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OpenStack / List Security Group

💾 Save

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GET

▾

http://192.168.27.10:9696/v2.0/security-groups

Send

▾

ParamsAuthorizationHeaders (8)BodyPre-request ScriptTestsSettingsCookies

Headers

👁 6 hidden

	Key	Value	Description	⋮	Bulk Edit	Presets ▾
☑	Content-Type	application/json				
☑	X-Auth-Token	{{openstack-token}}				
	Key	Value	Description			

BodyCookiesHeaders (7)Test Results

🌐 Status: 200 OK Time: 126 ms Size: 6.41 KB

💾 Save as Example

 ⋮

Pretty

Raw

Preview

Visualize

JSON ▾

☰

1

{

2

"security_groups": [

3

{

4

"id": "3ab8da26-1e48-437d-a39a-0cce1ff02ca",

5

"name": "baonv",

6

"stateful": true,

7

"tenant_id": "a069c5716c9d46deb99c9587322fbec9",

8

"description": "",

9

"shared": false,

10

"security_group_rules": [

11

{

12

"id": "4131610e-3cd7-439a-977a-b506506fc2f0",

13

"tenant_id": "a069c5716c9d46deb99c9587322fbec9",

14

"security_group_id": "3ab8da26-1e48-437d-a39a-0cce1ff02ca",

15

"ethertype": "IPv4",

16

"direction": "egress",

17

"protocol": "tcp"

Demo trực tiếp tại lớp

LIST SECURITY GROUP

Init-scripts -> Base64

```
#cloud-config
chpasswd:
  list: |
    root:root
  expire: False
  ssh_pwauth: True

I2Nsb3VklWNvbmZpZwpjaHBhc3N3ZDoKIGxpc3Q6IHwKICAgcm9vdDpyb290CmV4cGlyZTogRmFsc2UKc3NoX3B3YXV0aDogV
HJ1ZQ==
```

Demo trực tiếp tại lớp

TẠO INSTANCE

The screenshot shows the Postman interface with a POST request to `http://192.168.27.10:8774/v2.1/servers`. The request body is a JSON object defining an OpenStack server. The response status is 202 Accepted, with a time of 1166 ms and a size of 901 B. The response body is also shown in a pretty-printed JSON format.

Request:

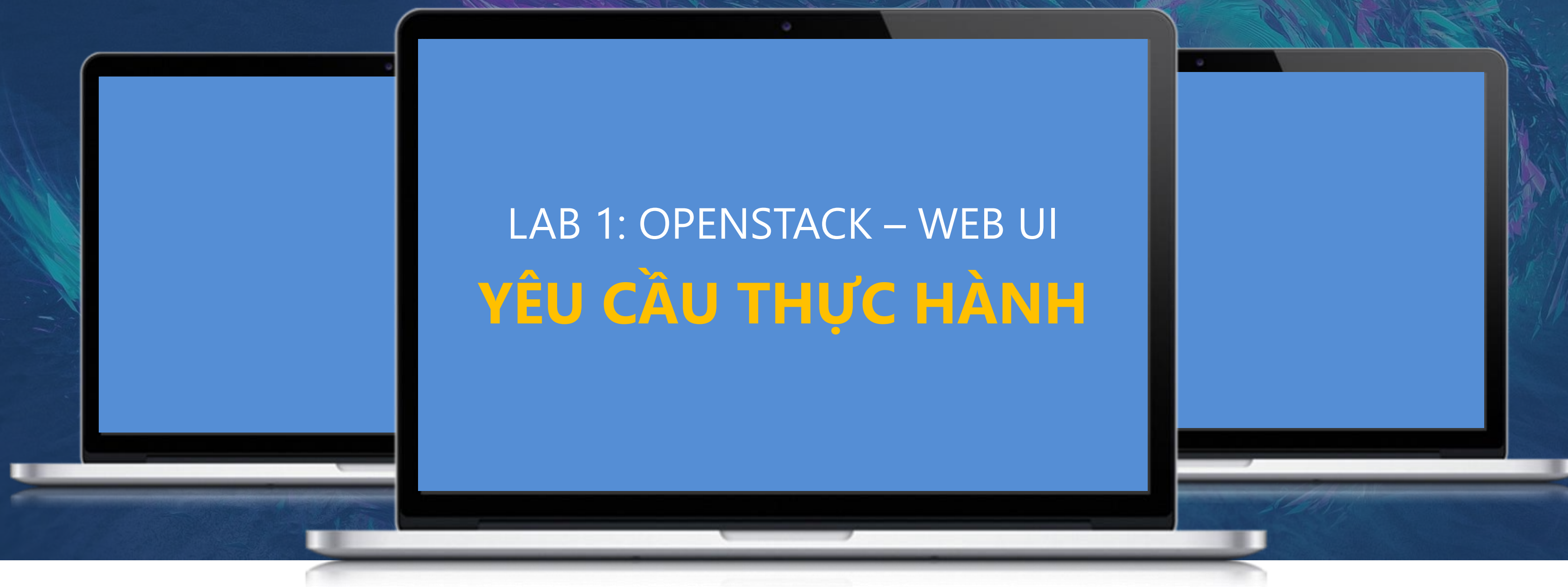
```
1 {
2   "server": {
3     "name": "nhomXX_PC1",
4     "imageRef": "819dc436-a8e1-4b08-8576-91ad1fc9292f",
5     "flavorRef": "02-02-45",
6     "security_groups": [
7       {
8         "name": "baonv"
9       }
10    ],
11    "networks": [
12      {
13        "uuid": "5bc9cf1a-5b07-4f99-950a-fd9fe24d8a1f",
14        "port": "2973459e-19cb-4c87-a0df-cef0cfbb4959"
15      }
16    ],
17    "user_data": "I2Nsb3VhbnVbmZpZw0KY2hwYXNzd2Q6DQogbG1zdDogfA0KICAgcm9vdDpyb290DQplcHBcmU6IEZhbHN1DQpzcz2hfchHdhXR0OiB0cnV1"
18  }
19 }
```

Response:

```
1 {
2   "server": {
3     "id": "71b853dd-31d3-46f6-bd40-311f6af18a11",
4     "name": "nhomXX_PC1",
5     "imageRef": "819dc436-a8e1-4b08-8576-91ad1fc9292f",
6     "flavorRef": "02-02-45",
7     "security_groups": [
8       {
9         "name": "baonv"
10      }
11    ],
12    "networks": [
13      {
14        "uuid": "5bc9cf1a-5b07-4f99-950a-fd9fe24d8a1f",
15        "port": "2973459e-19cb-4c87-a0df-cef0cfbb4959"
16      }
17    ],
18    "user_data": "I2Nsb3VhbnVbmZpZw0KY2hwYXNzd2Q6DQogbG1zdDogfA0KICAgcm9vdDpyb290DQplcHBcmU6IEZhbHN1DQpzcz2hfchHdhXR0OiB0cnV1"
19  }
20 }
```

Demo trực tiếp tại lớp

TẠO INSTANCE



Yêu cầu



Hướng dẫn



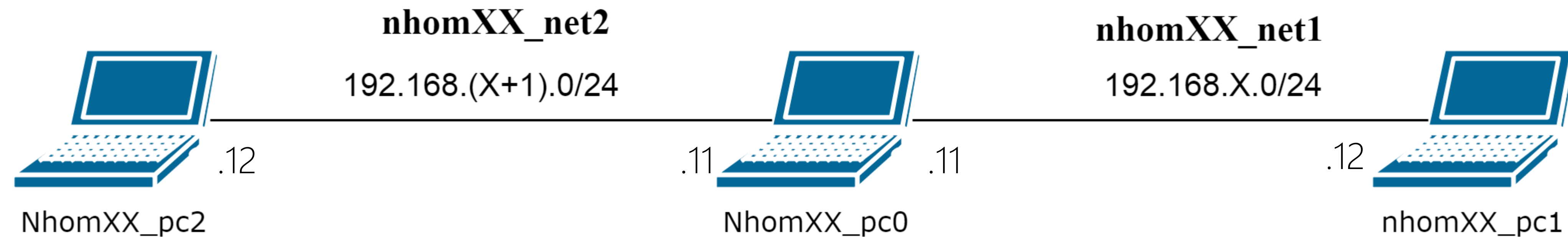
Báo cáo

YÊU CẦU 1

25

nội dung thực hành

Triển khai, cấu hình hệ thống gồm 3 máy tính kết nối theo kiến trúc mạng hình sau:



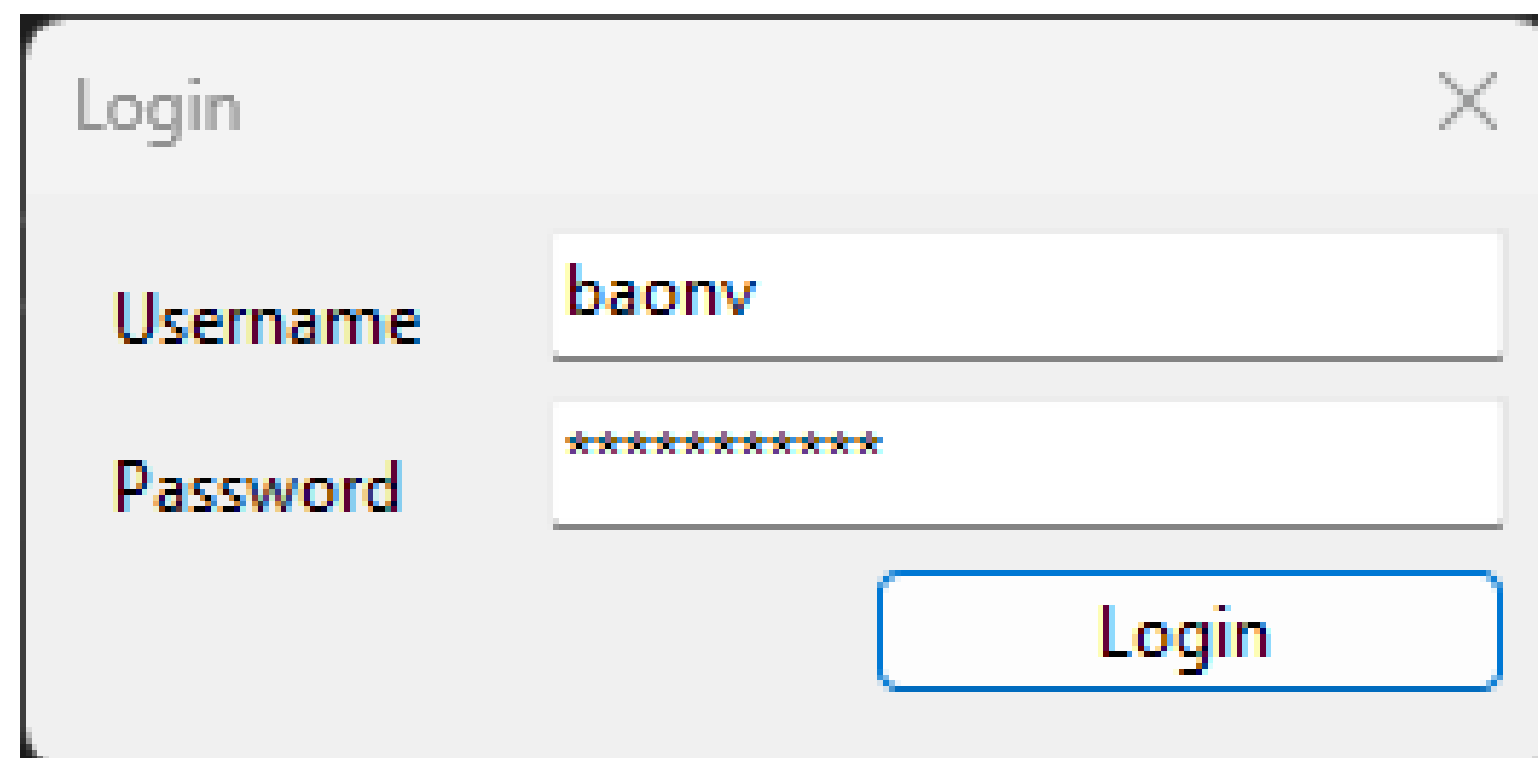
YÊU CẦU 2

26

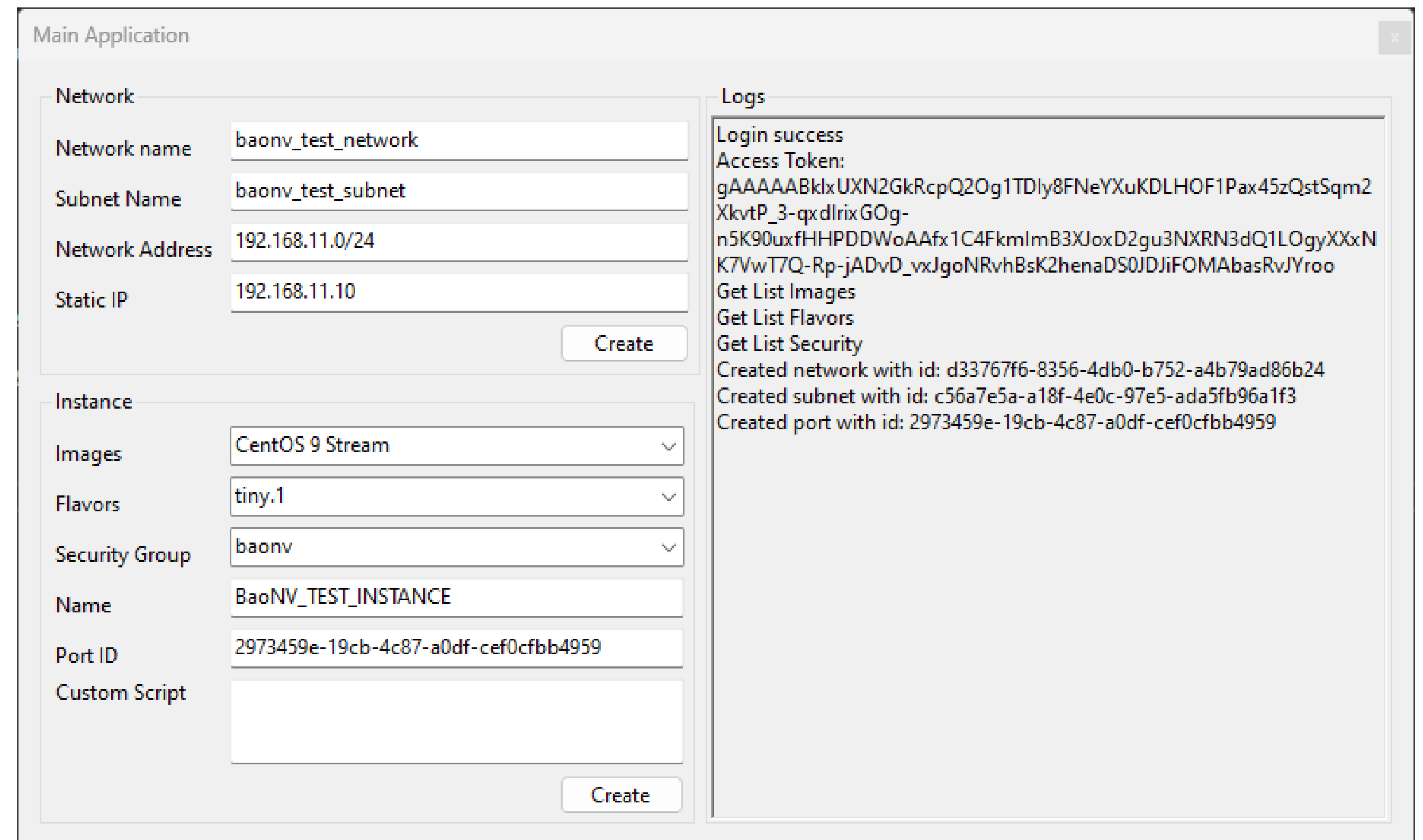
nội dung thực hành

- Viết phần mềm (Desktop app, web app, mobile app, ...) cung cấp giao diện cho người dùng có thể tương tác với hệ thống thông qua các API mà OpenStack cung cấp
- Sử dụng ngôn ngữ lập trình bất kỳ
- Giao diện tùy ý nhưng phải cung cấp đầy đủ các tính năng để có thể tạo được một instance

Ví dụ



A small login dialog box with a title bar 'Login' and a close button. It contains two input fields: 'Username' with the value 'baonv' and 'Password' with masked characters '*****'. A 'Login' button is at the bottom right.



A larger application window titled 'Main Application'. It has two main sections: 'Network' and 'Instance'. The 'Network' section has fields for 'Network name' (baonv_test_network), 'Subnet Name' (baonv_test_subnet), 'Network Address' (192.168.11.0/24), and 'Static IP' (192.168.11.10), with a 'Create' button. The 'Instance' section has dropdowns for 'Images' (CentOS 9 Stream), 'Flavors' (tiny.1), and 'Security Group' (baonv), and text fields for 'Name' (BaoNV_TEST_INSTANCE), 'Port ID' (2973459e-19cb-4c87-a0df-cef0cfbb4959), and 'Custom Script'. It also has a 'Create' button. On the right, a 'Logs' panel shows a list of messages: 'Login success', 'Access Token: gAAAAABkxUXN2GkRcpQ2Og1TDly8FNeYXuKDLHOF1Pax45zQstSqm2XkvtP_3-qxdlrixGOg-n5K90uxfHHPDDWoAAfx1C4FkmlmB3XJoxD2gu3NXRN3dQ1LOgyXXxNK7VwT7Q-Rp-jADvD_vxJgoNRvhBsK2henaDS0JDJiFOMAbasRvJYroo', 'Get List Images', 'Get List Flavors', 'Get List Security', 'Created network with id: d33767f6-8356-4db0-b752-a4b79ad86b24', 'Created subnet with id: c56a7e5a-a18f-4e0c-97e5-ada5fb96a1f3', and 'Created port with id: 2973459e-19cb-4c87-a0df-cef0cfbb4959'.

HƯỚNG DẪN

Các bước thực hiện

B1. Chứng thực

B2. Tạo các networks nhómXX_net1, nhómXX_net2

B3. Tạo các subnets nhómXX_subnet1, nhómXX_subnet2

B4. Tạo các ports: 192.168.X.11, 192.168.X.12, 192.168.(X+1).11, 192.168.(X+1).12

B5. List images

B6. List Flavors

B8. List keypairs

B9. Tạo instance (init-script -> base64 để tự động cấu hình instance)

B10. Kiểm tra

B11. Viết phần mềm.

B11. Viết báo cáo

BÁO CÁO

cần gì?

YÊU CẦU 1

Nội dung	Url	Request	Tóm lược response

Giải thích cơ bản kèm hình chụp

Phần mềm

Giải thích cơ bản kèm hình chụp



Now It's your turn!